Random Processes and Stationarity

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Key Concepts

1) A random process is a random signal $x[n]$, $-\infty < n < \infty$.

2) Complete probability descriptions for random processes are not normally used in signal processing.

3) In general the probability density function of a random process can be different for different times in the process.

4) A stationary process has the same characteristics (probability density function) at all points in time. For example:

a) The mean is a constant and does not depend on time.

b) The auto correlation and auto covariance between two elements of the random process don't depend on their absolute time locations, but only the time difference between them.

5) In a wide-sense stationary process the mean and auto covariance of the process are independent of time. That is, they are the same for all values of n.

6) A process is said to be ergodic if time averages are equal to ensemble averages (expected values).

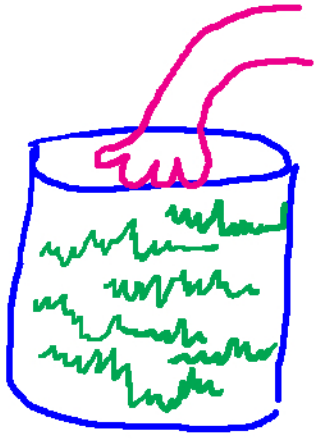
7) Statistics of a random process, such as mean and auto covariance, are normally estimated using time averages of a sample random process, i.e., a measured signal.

Video Screenshots

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Random process: $\{x[n],\ -\infty < n < \infty\}$

independent variable n denotes time



infinite vector $[\ldots x[-1], x[0], x[1], x[2], \ldots]$

covariance matrix $\underline{R}: N \times N,\ N \to \infty$?

Complete probability description is possible (pdf), but often not practical

Mean: $\mu[n] = E\{x[n]\}$

Autocorrelation: $\eta[n, n-k] = E\{x[n]x[n-k]\}$ lag k

Autocovariance: $r[n, n-k] = E\{(x[n]-\mu[n])(x[n-k]-\mu[n-k])\}$

Stationary Process – characteristics don't change over time

Mean $\mu[n] = \mu$

Autocorrelation $\eta[n, n-k] = \eta[k]$

Autocovariance $r[n, n-k] = r[k]$

} don't depend on n, only on separation distance k



$E\{x[9]x[4]\} = \eta[9,4]$

$n=9, k=5$

$E\{x[50]x[45]\} = \eta[50,45]$

$n=50, k=5$

$E\{x[85]x[80]\} = \eta[85,80]$

$n=85, k=5$

Stationary: $\eta[9,4] = \eta[50,45] = \eta[85,80] = \eta[5]$

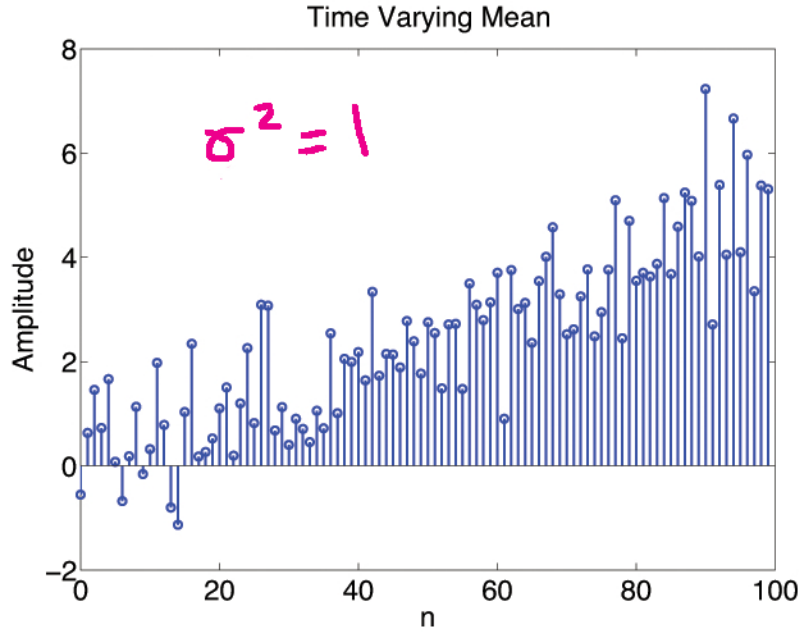
properties only depend on k

Examples of Nonstationary Signals

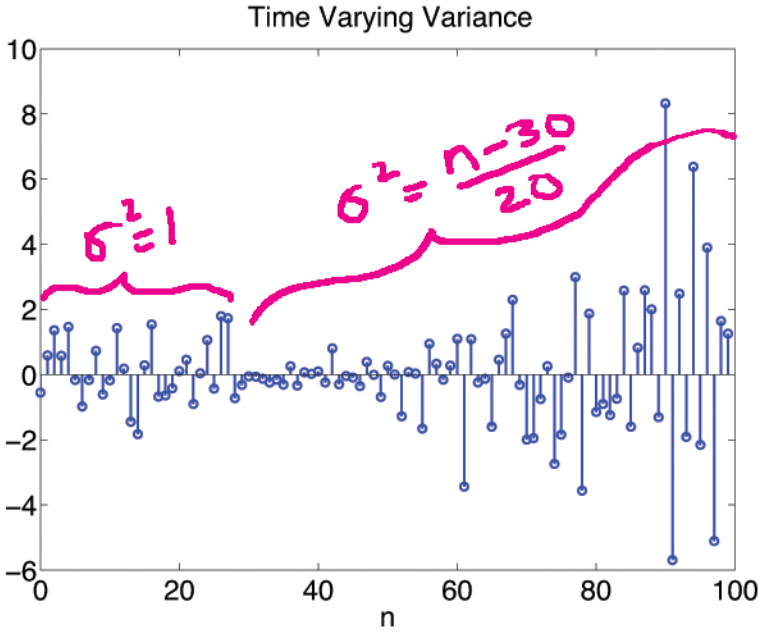
$A: \frac{r[1]}{r[0]} = 0.9$

$B: \frac{r[1]}{r[0]} = -0.9$

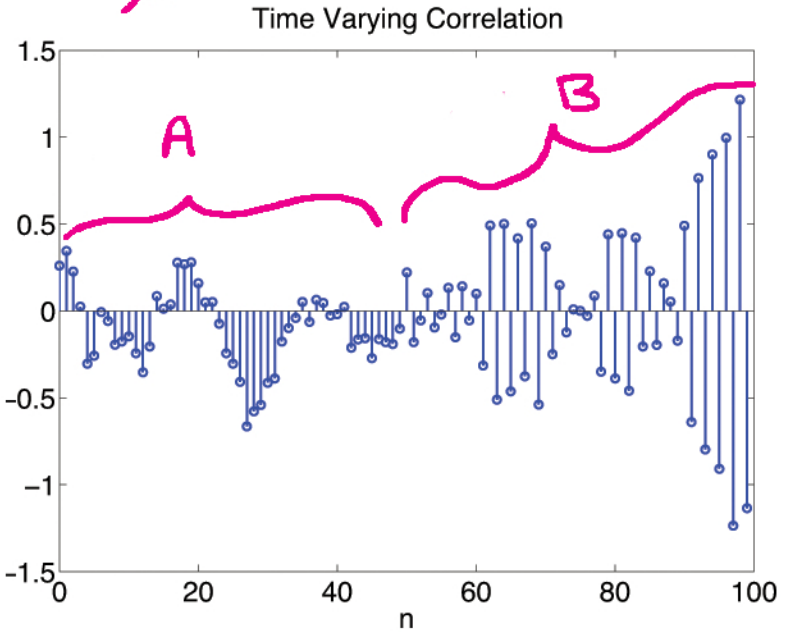
$\mu[n] = n/20$



$\mu = 0$



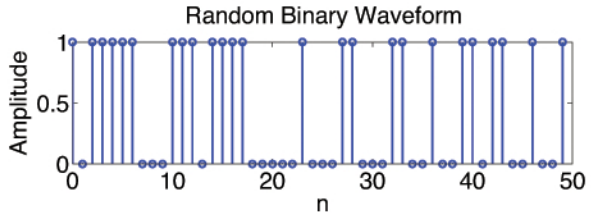
$\mu = 0$



Examples of Stationary Signals –

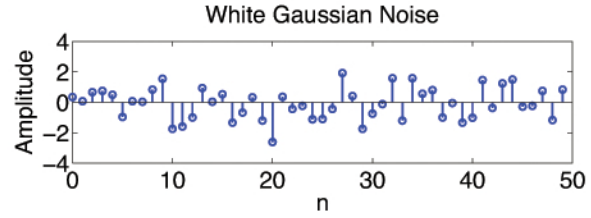
1) Random Binary Waveform $x[n] = \begin{cases} 0 & P = 1/2 \\ 1 & P = 1/2 \end{cases}$

$\mu = 1/2$, $\eta[k] = \begin{cases} 1/2 & k=0 \\ 1/4 & k \neq 0 \end{cases}$ $E\{x^2[n]\}$ $E\{x[n]x[n-k]\}$



2) White Gaussian Noise $w[n] \sim N(0, \sigma_w^2)$ independent samples

$\mu = 0$, $r[k] = \begin{cases} \sigma_w^2 & k=0 \\ 0 & k \neq 0 \end{cases}$

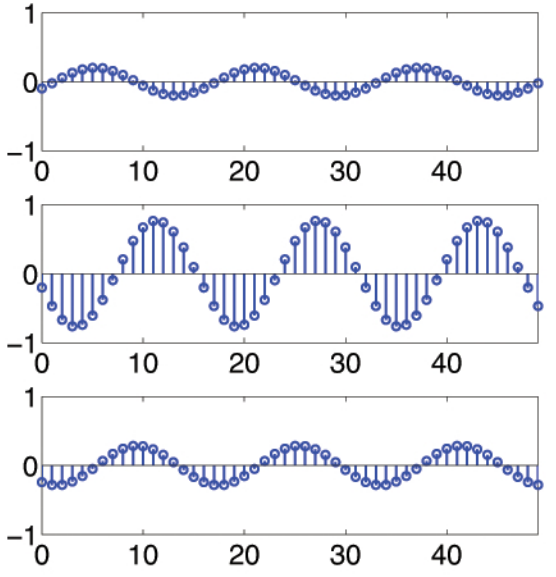


3) Random Sinusoid $y[n] = A\cos(\omega_0 n + \phi)$

$A \sim N(0, \sigma_A^2)$; $\phi \sim U(0, 2\pi)$

$\mu = 0$, $r[k] = E\{A^2 \cos(\omega_0 n + \phi)\cos(\omega_0(n-k) + \phi)\}$

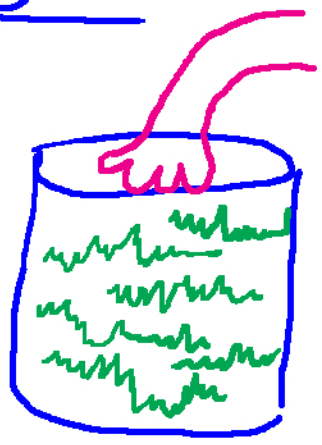
$= \frac{\sigma_A^2}{2}\cos(\omega_0 k)$



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Wide-sense stationary - mean, autocorrelation/autocovariance don't change with time

Ergodic - ensemble averages equal time averages



$$E\{x[n]\} = \mu \qquad N \text{ large} \qquad \mu = \frac{1}{N}\sum_{n=1}^{N} x[n]$$

$$\Leftrightarrow$$

$$E\{x[n]x[n-k]\} = \eta[k] \qquad \eta[k] = \frac{1}{N}\sum_{n=1}^{N} x[n]x[n-k]$$

In practice statistics are usually estimated from time averages

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